CSE321: Computer Networks
Class Test – 3

Time: 30 minutes **Mark: 20**

[Instructions:

1) This is an **OPEN-MATERIAL** online examination.

2) Write your answer in a sheet. Make sure you write your Student # there on top of the page(s).

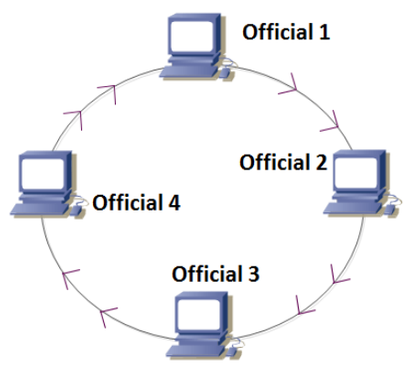
3) Then, make a pdf scanned copy of your answer and submit that (pdf) in Moodle *in time*.

4) Check and confirm at your own that your submitted file is okay and not corrupted.

Grading will be made based on what will be available in Moodle.]

<u>Question:</u>

A security agency needs to develop an extremely secured network among its top four officials. To do so, they have formed a ring topology consisting of the computing machines of the top officials as follows. In all the machines, only one security application is installed.



Here, in case of any data transmission, the above direction of flow is followed completing the full cycle. Thus, for example, if Official 1 wants to transmit a data, it sends data following the above direction reaching everyone in sequence of Official 1 -> Official 2 -> Official 3 -> Official 4 -> Official 1, eventually reaching own self and then terminate the transmission. The transmission does not care about any error control, flow control, or whatsoever while receiving own transmission as the final receiver. The transmission also does not care about any error control, flow control, or whatsoever while initiating transmission targeting the full ring traversal completion. However, each transmission cares about successful reaching to the next official's computing machine.

For the purpose of redundancy, the agency deploys two such ring topologies – one with fiber optic and another with copper cable. In the case of the fiber optic, digital transmission is enabled ensuring that a large number of harmonics get passed over the medium. On the other hand, in the case of the copper cable, analog transmission is enabled ensuring that a selective harmonic gets passed over the medium.

Now, for both the cases (fiber optic network and copper cable network), you need to sort out two protocol stacks so that the stacks contain optimal number of layers from the OSI reference model. For the purpose of optimization, make sure that you do not adopt any layer having minimal or no operation involved with it.

You need to write names of the adopted layers in your protocol stacks in order following the top to bottom hierarchy. Besides, you need to justify your answer on formation of your protocol stacks.